22 Here are the first five terms of a sequence.

4 11 22 37 56

Find an expression, in terms of n, for the nth term of this sequence.

4 11 22 37 56

+7 +11 +15 +19

+4 +4 +4

Second difference is 4, so the quadratic term is $2n^2$.

1 mark
Constant second difference: use a quadratic

$2n^2$: 2, 8, 18, 32, 50

Subtract: 2, 3, 4, 5, 6 = n + 1

1 mark
Find $2n^2$ and the remaining linear pattern

1 mark
Combine both parts

$2n^2 + n + 1$

(Total for Question 22 is 3 marks)